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1- OVERVIEW

The Blower Box (MG6) contains two blowers: One provides cooling air to the gradient body coil, and the other provides cooling air to the patient comfort module. The blowers are AC induction motors, so they do not have brushes. The entire motor must be replaced.

WARNING!

STRONG MAGNETIC FIELD! WHEN SERVICING ANY MAGNETIC EQUIPMENT, IT IS CRITICALLY IMPORTANT THAT THE SERVICE ENGINEER CONSCIOUSLY PLAN THE PATH TO BE TAKEN WHEN MOVING HIGHLY FERROUS DEVICES IN THE MAGNET ENVIRONMENT. THE PATH SHOULD BE AS FAR FROM THE MAGNET AS PRACTICAL.

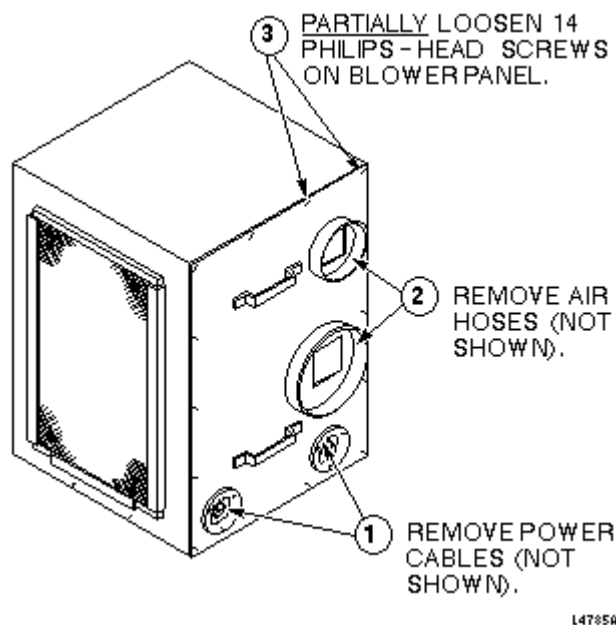
SAFETY REQUIREMENTS

- **MOVEMENT OF FERROUS MATERIAL IN THE MAGNET ROOM MUST FOLLOW THE GE SERVICE PROCEDURE FOR THAT DEVICE. WHEN EXITING, MOVE AWAY FROM THE MAGNET IN THE MOST DIRECT MANNER POSSIBLE. EXCEPT WHEN MOVING FERROUS MATERIAL TO AND FROM ITS NATIVE LOCATION ON OR NEAR THE MAGNET, THE STATIC MAGNETIC FIELD IN ANY PORTION OF THE SERVICE PATH SHALL NOT EXCEED 200 GAUSS.**
- **TWO (2) MR SAFETY TRAINED PERSONNEL MUST BE PRESENT AT ALL TIMES WHEN SERVICING HIGHLY FERROUS DEVICES IN THE AREAS OF MAGNETIC FIELDS.**

WHEN PLANNING A SERVICE PATH, IT IS CRITICAL THAT THE PATH BE CLEAR AND SUFFICIENTLY WIDE. ENSURE THAT THERE ARE NO TRIP HAZARDS, OBSTACLES, CLUTTER, SLIPPERY SURFACES OR OTHER ITEMS EVEN PARTIALLY RESTRICTING THE PATH. IF THERE ARE PORTABLE OBSTACLES IN A PATH, REMOVE THEM FROM THE AREA AND REPLACE THEM AFTER THE SERVICE ACTION IS COMPLETED. IT IS REQUIRED TO WALK THE PATH PRIOR TO BEGINNING SERVICE TO ENSURE THAT THERE IS SUFFICIENT SPACE THROUGH WHICH TO PASS FOR YOURSELF AND THE OBJECT BEING SERVICED.

2- REMOVING/INSTALLING BLOWER BOX MOTOR

1. Remove power to the Blower Box by switching circuit breakers MR1A14 CB1 (bore vent fan) and CB2 (gradient fan) on the rear of the MEPS module to the Off position in the **RF/Pen Cabinet**, or circuit breakers MR1A20 CB1 (bore vent fan) and CB2 (gradient fan) on the rear of the SSM to the Off position in the **RF/Pen II** and **RF/PDU Cabinet**.
2. Lock-Out / Tag-Out (LOTO) circuit breakers. Refer to the Safety procedure, *Section 6 - OSHA Lock-Out/Tag-Out Requirements* (SYSMSA2.DOC).
3. Verify that power is off by removing the two power cables at the Blower Box. (See Illustration 1-1.)



REMOVING BLOWER BOX MOTOR PANEL
ILLUSTRATION 1-1

4. Loosen the Blower Box Motor Panel from blower box shown in Illustration 1-1.



FERROUS MATERIAL HAZARD! TO PREVENT BODILY INJURY OR MATERIAL DAMAGE, STAY OUTSIDE THE 200 GAUSS LINE WHEN MOVING THE BLOWER ASSEMBLY. ANY FIELD STRENGTH >200 GAUSS WILL FORCIBLY ATTRACT THE MOTOR PANEL ASSEMBLY. TWO (2) MR SAFETY TRAINED PERSONNEL MUST PARTICIPATE IN THE REMOVAL AND REINSTALLATION OF THE MOTOR PANEL FROM THE MAGNET ROOM.

WARNING!

PINCH HAZARD AND POSSIBLE PERSONAL INJURY! DO NOT PLACE ANY PART OF THE HUMAN BODY IN THE PATH BETWEEN THE BLOWER BOX MOTOR PANEL ASSEMBLY AND THE MAGNET.

5. With two people holding the handles, slide the blower box motor panel assembly up and pull it slightly away from the blower box. (See Illustration 2-2.)



REMOVING BLOWER BOX MOTOR ASSEMBLY
ILLUSTRATION 2-2

6. With two people holding the motor panel assembly, exit the Magnet Room by walking as close to the wall on the coldhead side of the magnet as possible. (See illustration below.)

Note

To ensure a safe distance, make sure that the clear path through which you will be moving the blower box is completely outside the 200 Gauss line. The distance to the 200G line can be found in the Pre-Installation Manual (PIM).

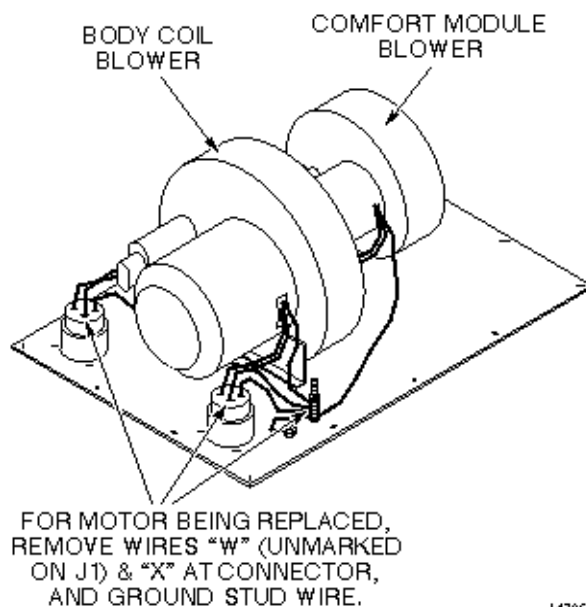
Important

The 200G line may **not** be within the minimum service area. If there is not enough room to move the blower box outside of the 200G line, the magnet must be ramped down **prior** to continuing.



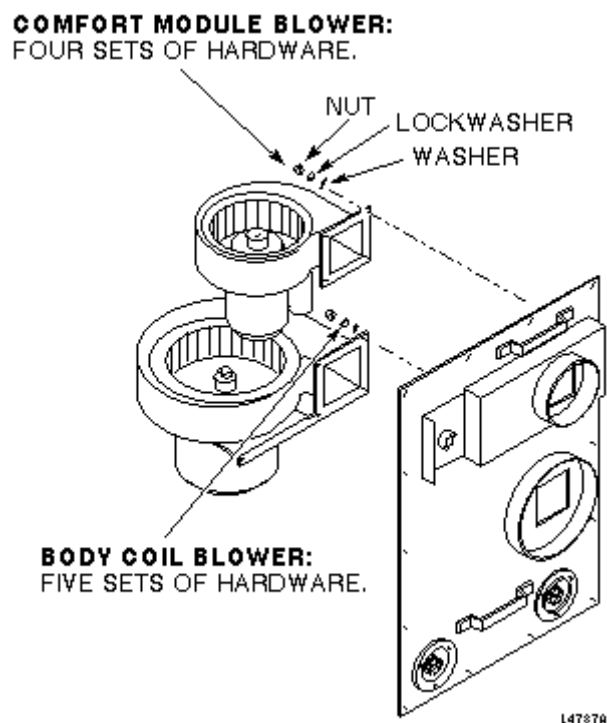
EXITING MAGNET ROOM WITH BLOWER BOX
ILLUSTRATION 2-3

7. Disconnect wiring from failed motor. (See Illustration 2-4.)



REMOVING BLOWER BOX MOTOR WIRING
ILLUSTRATION 2-4

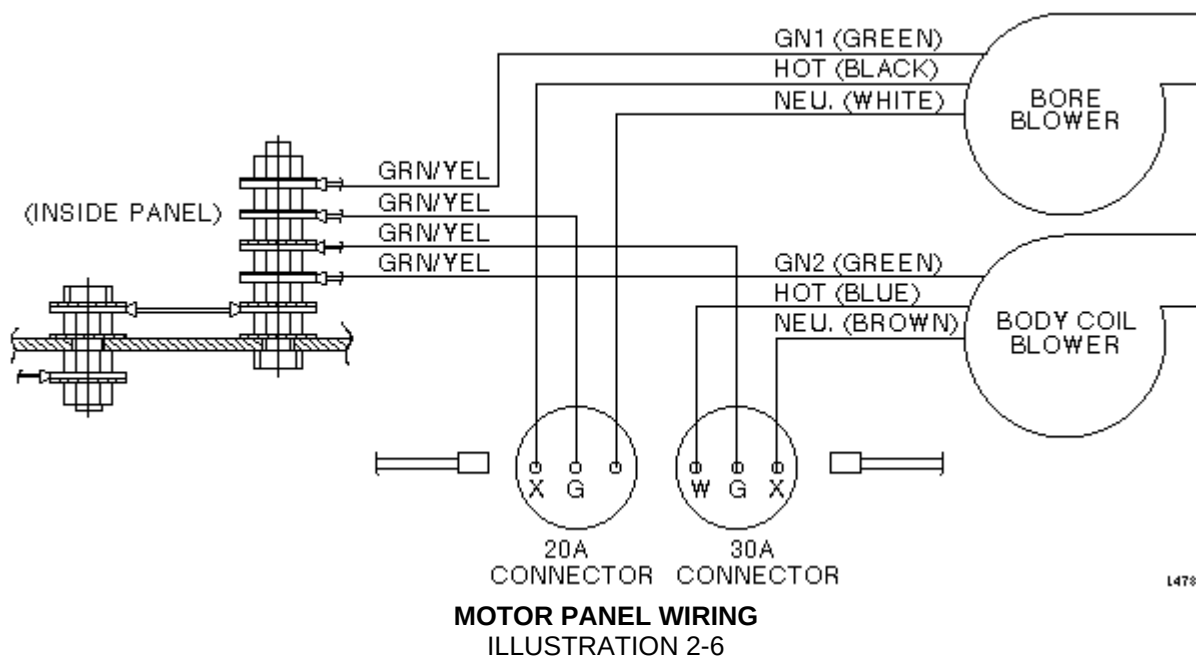
8. Remove the failed motor from the panel, and install the replacement motor. (See Illustration 2-5.)



REMOVING/INSTALLING BLOWER MOTOR
ILLUSTRATION 2-5

9. Reconnect the motor wiring. (See Illustration 2-6.) Strip 10 to 13 mm (3/8 to 1/2 in.) of the insulation from each wire.

- For HOT and NEU. (neutral) wires, inset the wire into the connector, and tighten the set screw.
- For GN1 or GN2 wires, terminate the wire with a 6 mm (1/2 in.) ring terminal, and connect to the ground stud.



DANGER!!

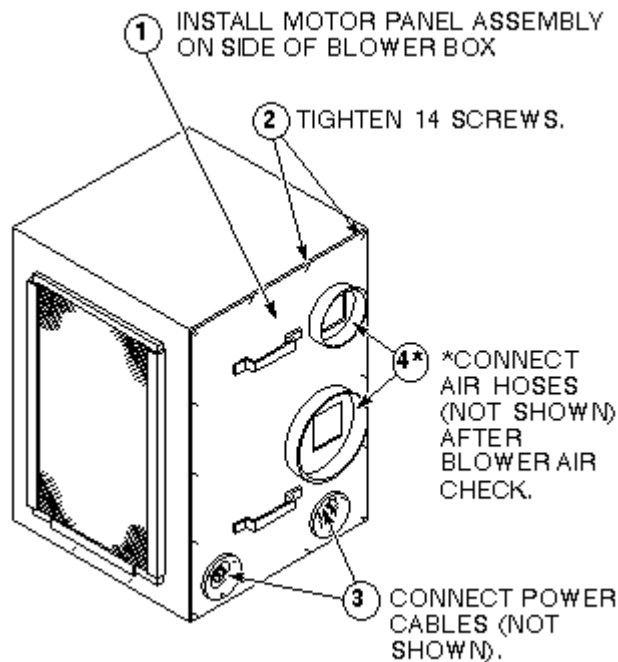
FERROUS MATERIAL HAZARD! TO PREVENT BODILY INJURY OR MATERIAL DAMAGE, STAY OUTSIDE THE 200 GAUSS LINE WHEN MOVING THE BLOWER ASSEMBLY. ANY FIELD STRENGTH >200 GAUSS WILL FORCIBLY ATTRACT THE MOTOR PANEL ASSEMBLY. TWO (2) MR SAFETY TRAINED PERSONNEL MUST PARTICIPATE IN THE REMOVAL AND REINSTALLATION OF THE MOTOR PANEL FROM THE MAGNET ROOM.

WARNING!

PINCH HAZARD AND POSSIBLE PERSONAL INJURY! DO NOT PLACE ANY PART OF THE HUMAN BODY IN THE PATH BETWEEN THE BLOWER BOX MOTOR PANEL ASSEMBLY AND THE MAGNET.

3. With two people holding the motor panel assembly, stay as far away from the magnet as possible while bringing the motor panel assembly into the Magnet Room. Walk as close to the wall on the coldhead side of the magnet as possible until reaching the blower box.

4. Reinstall the Motor Panel Assembly and power cables; wait to install the hoses until after the blower check. (See Illustration 2-7.)



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INSTALLING MOTOR PANEL
ILLUSTRATION 2-7

5. Remove LOTO from circuit breaker MR1A14 CB1 and CB2 at the rear of the MEPS module located in the rear of **RF/Pen Cabinet**, or remove LOTO from circuit breakers MR1A20 CB1 (bore vent fan) and CB2 (gradient fan) on the rear of the SSM in the **RF/Pen II** and **RF/PDU Cabinet**.
6. Switch circuit breakers MR1A14 CB1 and CB2 to the On position in the **RF/Pen Cabinet**, or switch circuit breakers MR1A20 CB1 (bore vent fan) and CB2 (gradient fan) on the rear of the SSM to the On position in the **RF/Pen II** and **RF/PDU Cabinet**.
7. Verify that both blowers are operating and blowing air out of the Blower Box.
8. Switch circuit breakers MR1A14 CB1 and CB2 to the Off position in the **RF/Pen Cabinet**, or switch circuit breakers MR1A20 CB1 (bore vent fan) and CB2 (gradient fan) on the rear of the SSM to the Off position in the **RF/Pen II** and **RF/PDU Cabinet**.
9. LOTO circuit breakers. Refer to the Safety procedure, *Section 6 - OSHA Lock-Out/Tag-Out Requirements* (SYSMSA2.DOC).
18. Reinstall the Blower Box hoses. (See Illustration 2-7.)
19. Switch circuit breakers MR1A14 CB1 and CB2 to the On position in the **RF/Pen Cabinet**, or switch circuit breakers MR1A20 CB1 (bore vent fan) and CB2 (gradient fan) on the SSM to the On position in the **RF/Pen II** and **RF/PDU Cabinet**.

3- ADJUSTMENT OF COMFORT MODULE (PATIENT) BLOWER AIRFLOW

An adjustment is provided to regulate the airflow for patient comfort. This adjustment flap may be set between the 1/4 and full open position based on ambient room temperature and the customer's preference. The full open position provides maximum airflow to the patient.

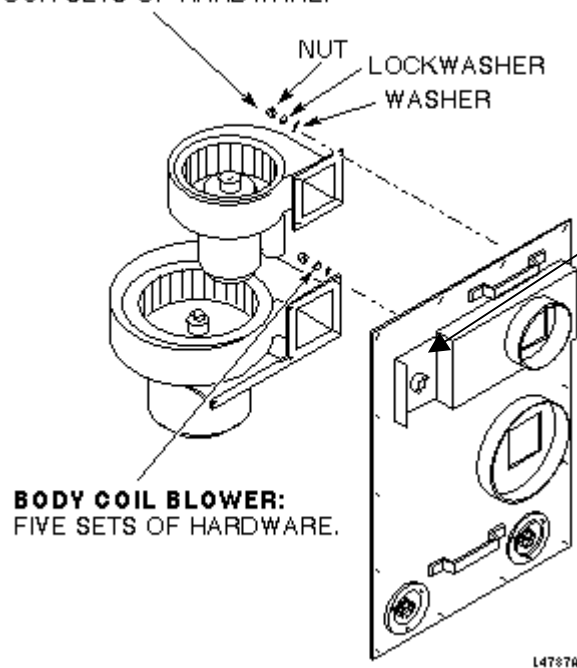
To adjust the airflow, refer to Illustration 3-1:

1. Loosen the nut, and slide the shutter to the desired position.
2. Tighten the nut to hold the shutter securely in the desired position.

Note

The shutter is normally set to the 1/2 open position. If the ambient room temperature is within the normal limits, the 1/2 open setting provides adequate airflow for most patients.

COMFORT MODULE BLOWER:
FOUR SETS OF HARDWARE.



COMFORT MODULE BLOWER ADJUSTMENT
ILLUSTRATION 3-1

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	April 27, 1998	W. Pasciak	Initial release converted to Word.
0	May 11, 1998	wek	Update lockout/tagout copy.
1	January 26, 1999	K. Keshena	Added information for the RF/PDU Cabinet.
2	May 17, 2004	R. Stetz	Section 3 Patient Blower Air Flow Adjustment.
3	Sept. 15, 2010	R. Kohl	Updated per CAPA 1859392: • Section 1 Overview - Added Warning. • Section 2 Removing/Installing Blower Box Motor - Updated procedure, Warnings and added pinch hazard Warning.